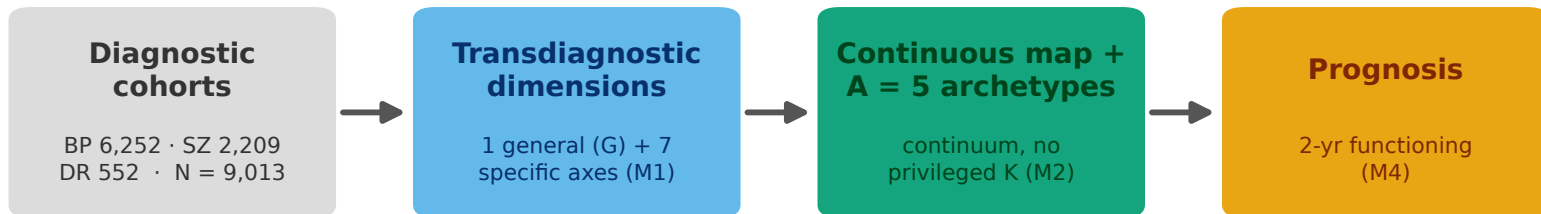


A latent atlas of severe mental illness



diagnosis enters only as a held-out validation label — never as a model input

Three load-bearing invariants

- ✓ No imputation — the model is fit to each patient's observed cells only (mean cell missingness 39.8%, preserved).
- ✓ Diagnosis is metadata — cohort and DSM-5 subtype are covariates / validation labels, never indicators.
- ✓ Baseline (V0) defines the map — the 12- and 24-month visits validate it; no discovery on follow-up.

Four questions, increasingly hard:

exists

organizes

persists

predicts